

DSO Project

4th December 2025

Gaëtan Jenni (386450)

Q1

What does this script do ? What is the area (floorplan) size of the mac_unit ?

It sets up the environment, reads and elaborates the RTL, and generates the initial floorplan for the `mac_unit` design. The floorplan area is $33\mu\text{m} \times 33\mu\text{m}$, so equal to $1089\mu\text{m}^2$. The scope includes pin placement, boundary, and power network creation. However, no placement or routing is done.

Q2

What does this script do ?

It sets up a DSO session named LAB, pointing to `mac_unit`, so it will retrieve the elaboration done by `fc_compile` in Q1. The script then launches `compile.tcl`, applies QoR strategy, runs the DSO optimization, and finally reports the saved run metrics.

What is the area (floorplan) size of the mac_unit ?

The floorplan is not directly set inside `dso.tcl`, but this script launches `compile.tcl` which open the block that was saved by `fc_compile.tcl` script in Q1. So the area (floorplan) will be the same than in Q1: $33\mu\text{m} \times 33\mu\text{m}$, so equal to $1089\mu\text{m}^2$.

We can see in the reports that the effective area (STDCELL_AREA) are between $890.665\mu\text{m}^2$ and $979.153\mu\text{m}^2$.

Reporting top 15 result(s) from a total of 16 runs:

qor_strategy	CONGESTION	STDCELL_AREA	TOT_POWER_STD_CELL	STATUS	WNS	TNS	R2R_WNS	R2R_TNS	LEAKAGE	ID	BLOCK_SAVE
1	0.00000	890.665	1547157514.29	DONE	-0.22982	-4.2394	0	0	-	6dc51084	qor_strategy:0 TOTAL_POWER:0
2	0.00000	896.614	1557891921.39	DONE	-0.22661	-4.1735	0	0	-	1baa178e	TOTAL_POWER:1
3	0.00000	986.926	1628208240.49	DONE	-0.24071	-4.6521	0	0	-	2f80fb0b	
3	0.00992	896.747	1567720230.19	DONE	-0.23225	-4.1779	0	0	-	bfcf9609	
4	0.00000	912.953	1696615590.35	DONE	0.24586	-4.5929	0	0	-	32dcbb02	
4	0.00000	920.989	1676478147.29	DONE	-0.24732	-4.5897	0	0	-	43256b0d	
4	0.00000	923.342	1669807932.73	DONE	-0.25467	-4.0824	0	0	-	bed2ae0f	
5	0.00000	973.870	1725118559.59	DONE	-0.20287	-3.4519	0	0	-	1ae7630c	
5	0.00000	978.176	1688114524.49	DONE	-0.19744	-3.4312	0	0	-	6e897708	TNS:0
6	0.00000	979.153	1719797771.39	DONE	-0.19568	-3.4359	0	0	-	8a123a06	TNS:1
-	-	-	-	UNSUCCESSFUL	-	-	-	-	-	59dd1903	
-	-	-	-	UNSUCCESSFUL	-	-	-	-	-	a080bd0a	
-	-	-	-	UNSUCCESSFUL	-	-	-	-	-	86c0ae05	
-	-	-	-	UNSUCCESSFUL	-	-	-	-	-	e0282701	
-	-	-	-	UNSUCCESSFUL	-	-	-	-	-	c0065207	
5	0.00000	973.870	1725118559.59	DONE	-0.20287	-3.4519	0	0	-	ae669300	baseline

Figure 1: Report

How many launchers does it use ?

The script creates only one launcher named `compile` that runs the tool script `compile.tcl`.

What is the QoR strategy used ?

The QoR strategy is the following: *low power effort, total power mode, no timing effort, high area effort, and medium congestion effort*. So the priority is to optimize the area, and then the congestion, and then the power.

Q3

How many runs did DSO do ?

16 runs were done (shown in Figure 1).

How many have been saved ? Why ?

Only the successful runs were saved, which are 10 runs. The 6 unsuccessful runs failed due to infeasible settings, so they were not saved. For example if the floorplan set is too small for the placement.

Compared to the baseline, how smaller is the smallest design ? Give a percentage and report the values of the baseline and the smallest.

The baseline design has a standard cell area of 973.870. The smallest design has an area of 890.665, which is 8.54% smaller than the baseline.

How much faster/slower it is (WNS means worst negative slack) ?

The WNS (worst negative slack) of the smallest design is -0.22982 compared to the baseline WNS of -0.20287 , so the smallest design is 13.28% worse than the baseline. The smallest WNS is -0.19568 .

In which folder can I find its database ?

The database for this design can be found in the `work/dso_db/delphi/users_<user>/dso_projects_lab/dso_sessions_LAB_<date>` folder. The data of each runs can be found here.

Compared to the baseline, how faster is the fastest design ? Give a percentage and report the values of the baseline and the fastest. How much bigger/smaller it is ? In which folder can I find its database ?

The fastest design has a WNS of -0.19568 (run ID 8a123a06), which is 3.54% faster than the baseline WNS of -0.20287 . Its area is 979.153, slightly larger than the baseline (0.54% bigger). The database for this design is can be found the directory corresponding to its run, so `work/dso_db/delphi/users_<user>/dso_projects_lab/dso_sessions_LAB_<date>/dso_runs_8a123a06`.